Lecture 5: Omitted Variable Bias

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Case and Paxson (2008)

Representative sample of British 30-year-olds born in 1970

Run following regression (separately for men and women):

$$\log \text{Hourly Earnings}_i = \alpha + \beta \cdot \text{Height in Inches}_i + \varepsilon_i$$

Estimate $\hat{\beta} = 0.010$ for men, $\hat{\beta} = 0.015$ for women

- Man who is one inch taller will have ≈1% higher earnings

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Next, add controls for childhood IQ:

$$\log \text{Hourly Earnings}_i = \alpha + \beta_1 \cdot \text{Height in Inches}_i + \beta_2 \text{IQ at Age } 5_i + \beta_3 \text{IQ at Age } 10_i + \varepsilon_i$$

Estimate $\hat{\beta}_1 = 0.004$ for men, $\hat{\beta}_1 = 0.006$ for women

- Adding further controls (e.g. for parent's socioeconomic background and conditions in childhood) cuts effect to near-zero and not statistically significant

Exercises

Exercise 1. Consider the linear model $Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$. For each of the following examples, think of (a) some variables that are likely to *causally* affect the outcome, and (b) some variables that are likely to be *correlated* with the regressor.

- **Fiscal Multipliers:** Y_t = Output Growth, X_t = Government Spending Growth
- **Returns to Job Training:** Y_i = Income, X_i = Participation in Job Retraining Program
- **Returns to Capital:** Y_i = Quarterly Output, X_i = Firm's Capital Stock
- **Healthcare:** Y_i = Indicator for Mortality, X_i = Went to Hospital in Last Month
- **Income-Mortality:** Y_i = Age at Death, X_i = Income at Age 40
- **Crime:** Y_i = Neighborhood Crime Rate, X_i = Police Officers per Capita

Exercise 2. For each example in the exercise above, are there any variables that satisfy both (a) and (b)? What is the likely sign of the omitted variable bias coming from that omitted variable?